GG-R-550B
April 17, 1981
SUPERSEDING
Fed. Spec. GG-R-550A
May 14, 1975

FEDERAL SPECIFICATION

RODS, STADIA: FOLDING, WITH CASE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types and one style of folding stadia rods.

1.2 Classification. The rods shall be the following types and styles, as specified (see 6.2):

Type II - Metric graduations.

Type III - Combination rod.
Style B - Metric decimal-foot graduations (metric inside).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Process for Cotton Cloths.

FSC 6675

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Federal Standards:

Fed. Std. No. H28	- Screw-Thread Standards for Federal Services.
Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).
Fed. Std. No. 595	- Colors.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-3530	- Thread and Twine, Mildew Resistant or Water Repellent Treated.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-C-45662	- Calibration System Requirements.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129	- Marking for Shipment and Storage.
MS15795	- Washer, Flat-Metal, Round, General Purpose (In/mm).
MS24665	- Pin, Cotter (Split).
MS35425	- Nut, Plain, Wing, UNC-2B.
MS35494	- Screw, Wood, Flat Head, Slotted, Steel and Brass.
MS35646	- Screw, Eye-Steel or Brass.
MS35754	- Bolt, Square Neck, Countersunk (Plow), Steel, Cadmium or Zinc Plated, UNC-2A.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)

ANSI/ASTM A 108	- Steel Bars, Carbon, Cold-Finished, Standard Quality.
ANSI/ASTM B 16	- Free-Cutting Brass Rod, Bar, and Shapes for use in Screw Machines.
ANSI/ASTM B 36	- Brass Plate, Sheet, Strip, and Rolled Bar.
ANSI/ASTM B 209	- Aluminum Alloy Sheet and Plate.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

D 2016	- Moisture Content of Wood.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The stadia rods shall be as shown on Figure 1 through 8, and as specified herein.

3.2 First article (first produced stadia rod). The contractor shall furnish one or more stadia rods as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce stadia rods that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Wood.

3.3.1.1 Hardwood. Hardwood shall be maple or basswood and shall be free of decay, splits, streaks, checks, shakes, wane, holes, warp, and bow.

3.3.1.2 Softwood. Softwood shall be Port Orford cedar, ponderosa pine, white pine, sugar pine, or spruce. Softwood shall be free of decay, splits, streaks, checks, shakes, wane, holes exceeding 1/16-inch in diameter, and

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knots exceeding 1/2-inch in average diameter. Holes shall be filled and knots shall be sound and tight. Knots in clusters will not be permitted. Slope of grain shall not exceed 1 inch in 10 inches in the wider dimension of the rod and shall not exceed 1 inch in 30 inches in the narrow dimension of the rod. The wood shall not be warped or bowed.

3.3.1.3 Moisture content. The moisture content of hardwood and softwood shall be not less than 7 percent nor more than 14 percent as determined in accordance with ASTM D 2016.

3.4 Screw threads. All screw threads shall conform to Fed. Std. No. H28.

3.5 Circular level vial. The bracket and level dish shall be made flush with the face of the rod. The level shall have a sensitivity of not more than 20 minutes of arc for a 2-millimeter displacement.

3.6 Locking pin. The locking pin with attached cord shall be constructed as shown in Figure 6. The cord shall be no longer than that required to place the locking pin through the hasp to lock the upper and lower sections of the rod. The cord shall be looped and fastened together securely to the screw eye and locking pin.

3.7 Tolerances. Unless otherwise specified (see 6.2), tolerances for the rod shall be plus or minus 0.01-inch on decimals, and plus or minus 1/64-inch on fractions. The measuring equipment shall be calibrated in accordance with MIL-C-45662.

3.8 Graduations and numbers. Graduations and numbers may be either painted, stamped or screened directly onto the treated surface of the rod or affixed permanently to a marking strip, as specified (see 6.2). When applicable, marking strips shall be plastic, metal or laminate, as specified (see 6.2). The marking strip shall be affixed permanently to the rod by an impervious glue or cement. Graduations and numbers shall be painted, screened or printed. The photographic process shall be applied to metal marking strips only.

3.9 Carrying case. The stadia rod shall be fitted into a case made from cotton duck conforming to CCC-C-419, Type III, minimum weight per square yard 8.25 ounces for the body of the case and 18.48 ounces for the reinforcing duck at the closed end. The reinforcing duck shall be one piece wrapped around the outside bottom of the case body. The open

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end shall be provided with a strap, flap and buckle to completely enclose and hold the sections. Webbing conforming to MIL-W-530, Type II, Class 4, width 3/4 inch; shall be used as binding around all edges. The cotton duck shall be dyed olive drab and treated for mildew resistance in accordance with CCC-D-950, Type II, Class B, Inhibitor (f).

3.9.1 Thread. The cotton thread used to fabricate the carrying case shall be rendered fungus resistant in accordance with MIL-T-3530, Type I, Class 1, and dyed olive drab.

3.10 Identification marking. The rod and case shall be identified with manufacturer's name and model number.

3.11 Treatment and painting. Treatment of wood surfaces shall be in accordance with MIL-T-704, Type A.

3.11.1 Type II metric graduations. Unless otherwise specified (see 6.2), the color of Type II metric graduations shall match FED. STD. No. 595, Color Numbers 37875 or 17038.

3.11.2 Type III combination graduations, Style B. Unless otherwise specified (see 6.2), the color of Type III combination graduations, Style B, shall be the same as the metric graduations.

3.12 Workmanship. All parts of the stadia rod shall be clean and free from warping, chipping, seaming, and splintering. All edges shall be rounded or beveled. All fittings shall be securely mounted and all screws shall be fully turned down. When folded, the rod shall fold flat with both sections in alinement. Hinge joints shall be rigid when extended and locked. The perimeters of the graduations and markings shall be clean and shall show no extraneous or overlapping paint.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced stadia rod inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First produced stadia rod inspection.

4.3.1 Examination. The first produced stadia rod shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first produced stadia rod shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The stadia rod shall be examined as specified herein for the following defects:

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Major

101. Material not as specified.
102. Circular level not as specified.
103. Locking pin not as specified.
104. Screw threads not as specified.
105. Tolerances not as specified.
106. Measuring equipment not calibrated as specified.
107. Graduations not as specified.
108. Carrying case not as specified.
109. Colors not as specified.
110. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5.2 Test.

4.5.2.1 Level vial sensitivity. The enclosed vial shall be placed on a leveled beam, one end of which is fitted with a micrometer screw to provide a known angular movement of the beam with each revolution of the screw. Using a scale, measure the bubble displacement after moving the beam through an arc of 20 minutes. Bubble displacement of more than 2 millimeters shall constitute failure of this test. When specified (see 6.2), the contractor may certify compliance in lieu of testing.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing and marking shall be examined to determine compliance with the Quality Assurance Provisions of MIL-S-3534.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. The stadia rods shall be preserved and packaged in accordance with level A preservation and packaging requirements of MIL-S-3534.

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5.1.2 Level B. The stadia rods shall be preserved and packaged in accordance with the level B preservation and packaging requirements of MIL-S-3534.

5.1.3 Commercial. The stadia rods shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the stadia rods against damage during multiple shipments, handling and storage.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The stadia rods shall be packed in accordance with the level A packing requirements of MIL-S-3534.

5.2.2 Level B. The stadia rods preserved and packaged shall be packed in accordance with the level B packing requirements of MIL-S-3534.

5.2.3 Commercial. The stadia rods, preserved and packaged as specified in 5.1, shall be packed in fiberboard boxes to insure delivery to destination; provide for redistribution by the initial receiving activity; and be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification rules.

5.3 Marking.

5.3.1 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The stadia rod is a component of survey equipment for use with a surveying transit or alidade to measure ground distances and elevations in making topographic surveys.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and style of stadia rod required (see 1.2).
(c) Time frame required for submission of first produced stadia rod and number of stadia rods required (see 3.2).
(d) Tolerances if other than as specified (see 3.7).
(e) Method of placing graduations and numbers on stadia rods (see 3.8).
(f) Type of material required when marking strips are specified (see 3.8).
(g) When color of graduations shall be other than as specified (see 3.11.1 and 3.11.2).
(h) When the contractor may certify compliance in lieu of testing (see 4.5.2.1).
(i) Degree of preservation and packaging and degree of packing required (see 5.1 and 5.2).

6.3 First produced stadia rod. Any changes or deviations of production stadia rods from the approved first produced stadia rod during production will be subject to the approval of the contracting officer. Approval of the first produced stadia rod will not relieve the contractor of his obligation to furnish stadia rods conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Classification change. Type I, decimal-foot graduation, and Type III, Style A, decimal-foot-metric graduations, are discontinued.

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Preparing activity:

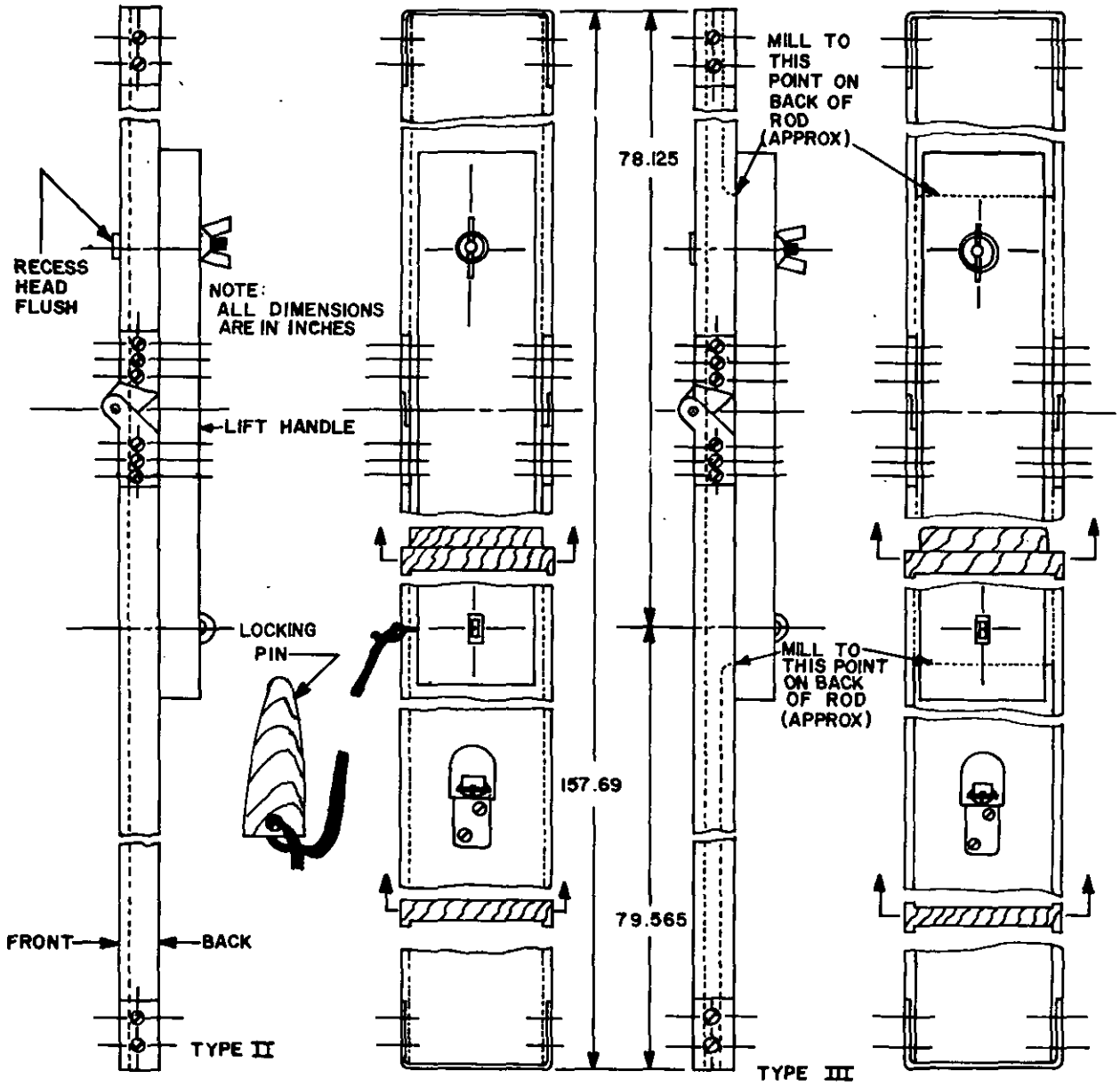
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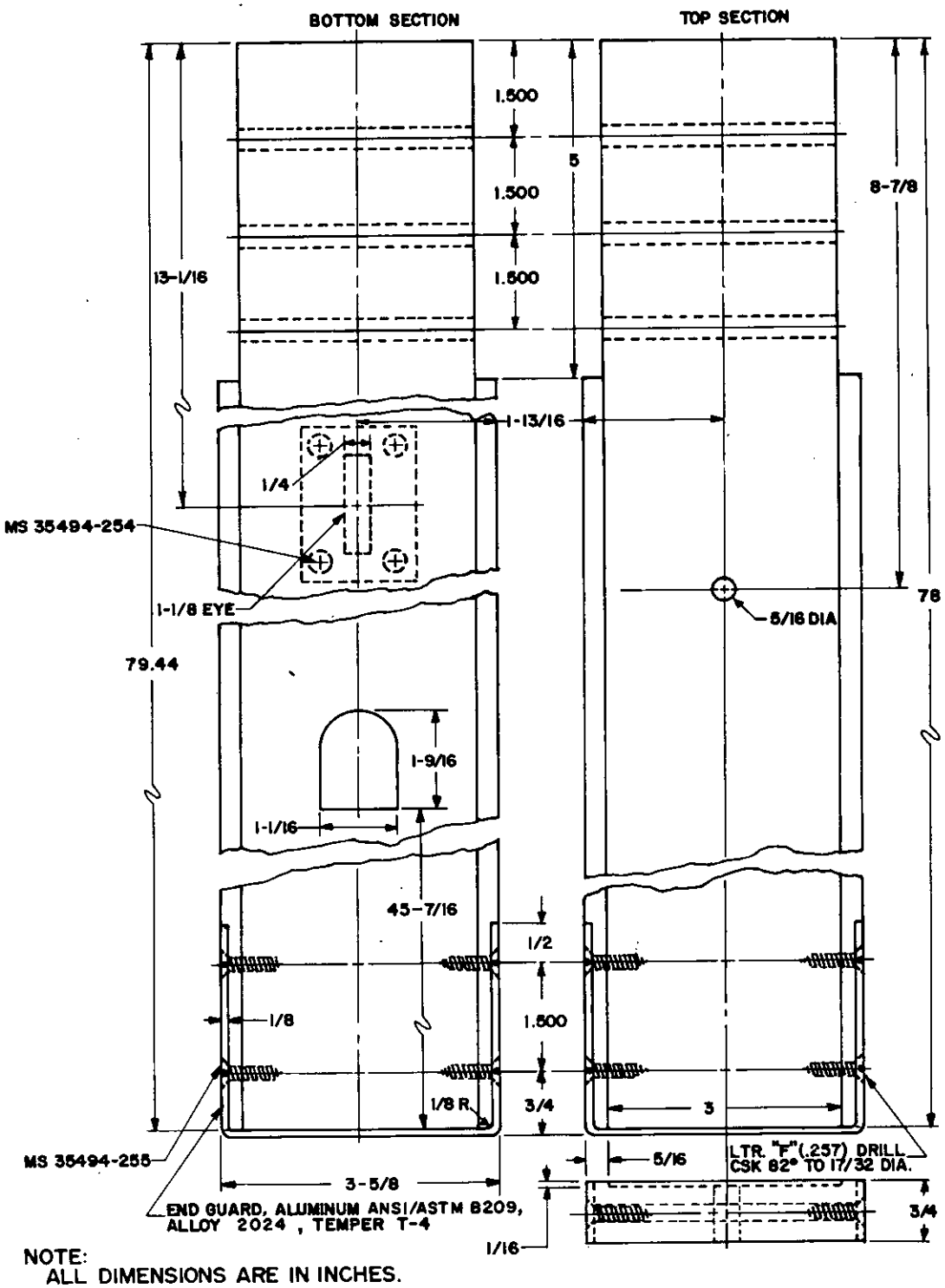
TYPE II - MILL 1/16 INCH ENTIRE FACE LENGTH

TYPES III - MILL 1/16 INCH ENTIRE FACE LENGTH AND ONLY TO THE POINT INDICATED ON BACK OF ROD

FIGURE 1. Stadia rod assembly.

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NOTE:
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FIGURE 2. Rod section details.

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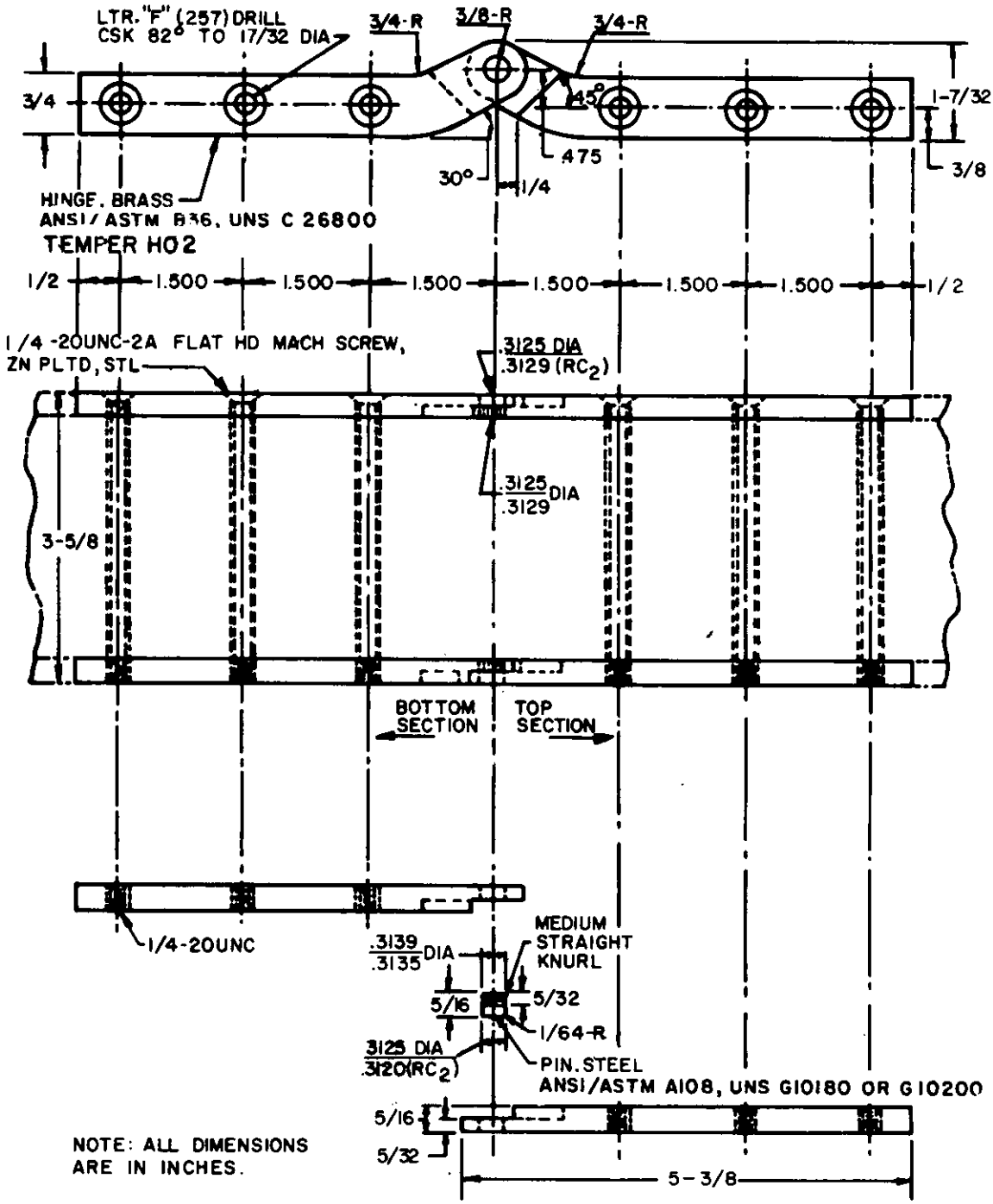


FIGURE 3. Hinge details.

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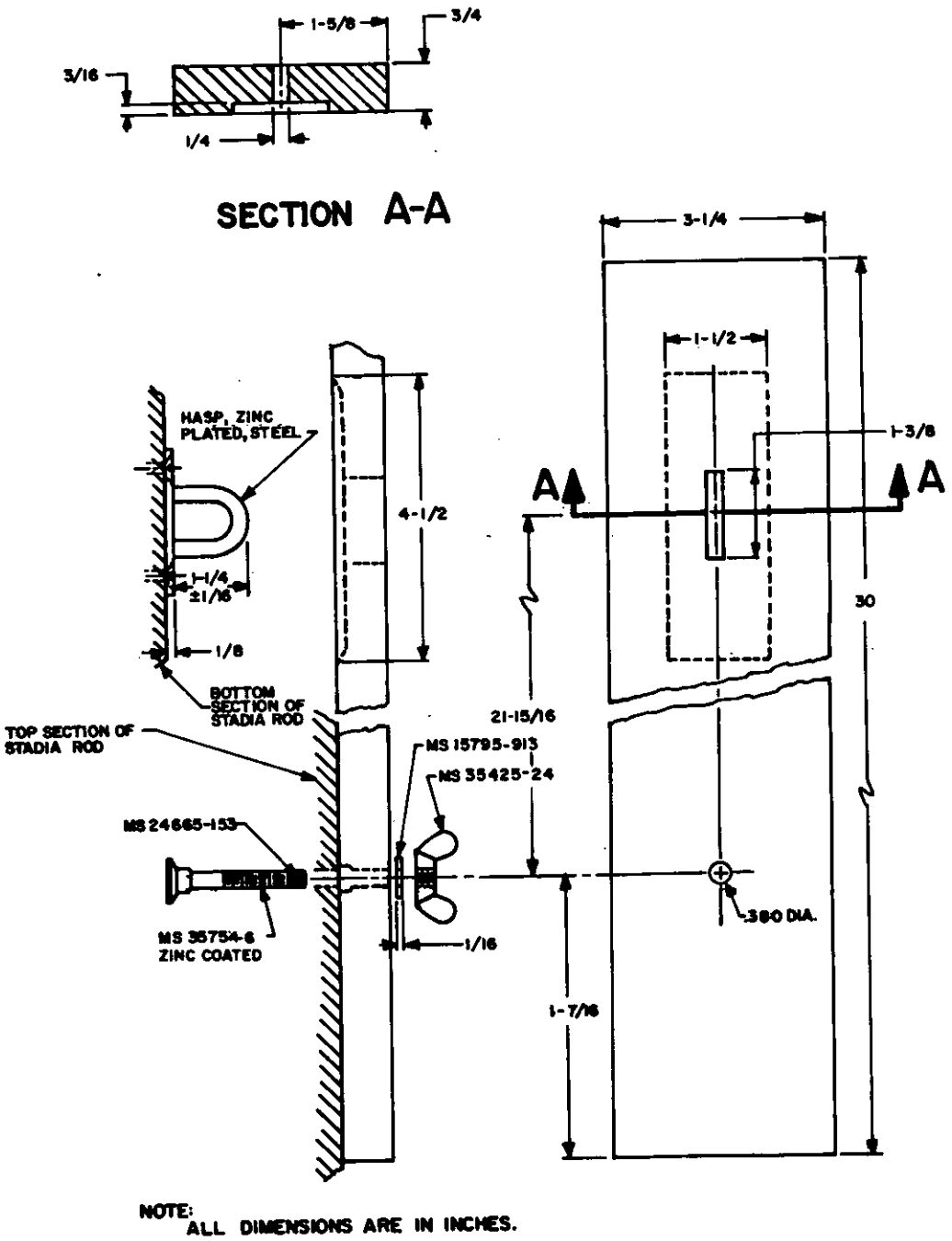
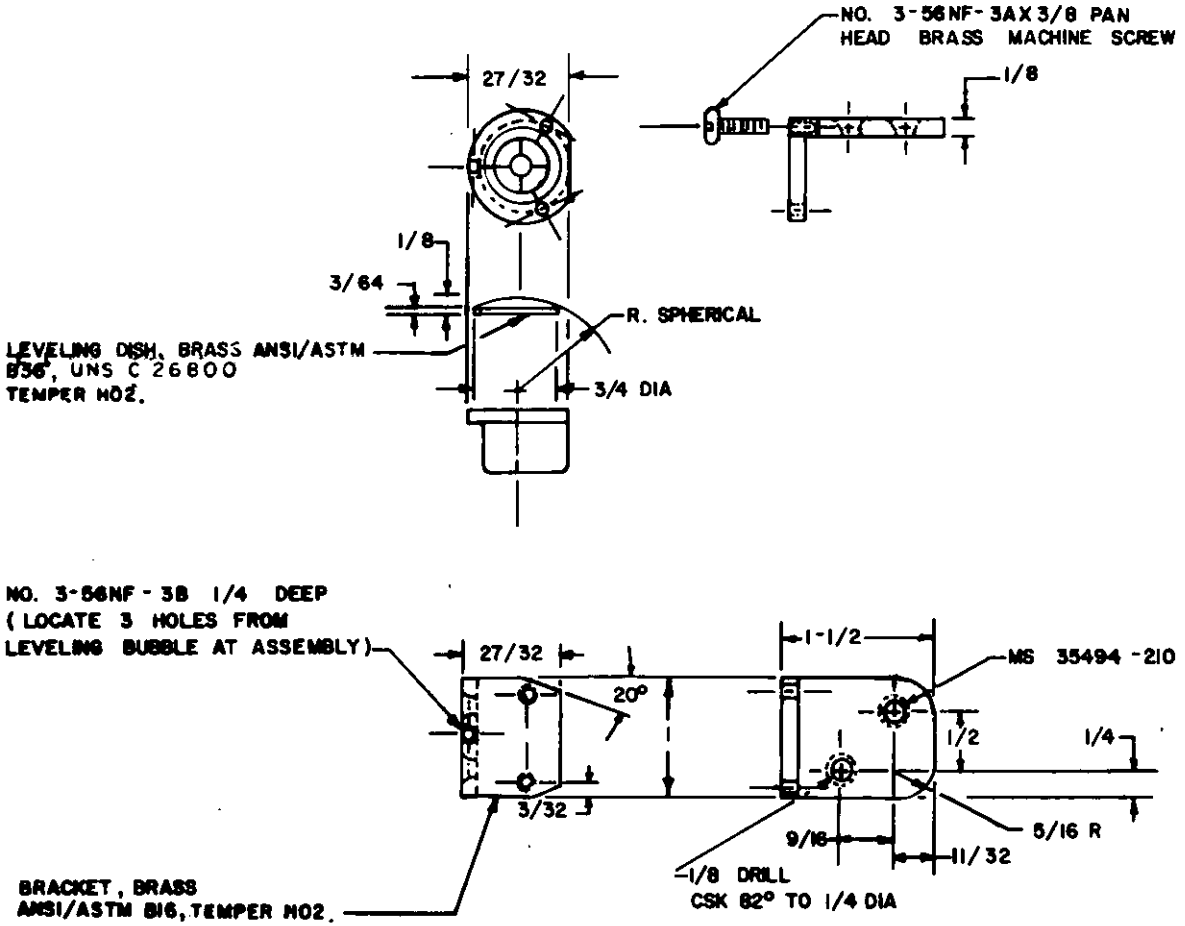


FIGURE 4. Lift handle details.

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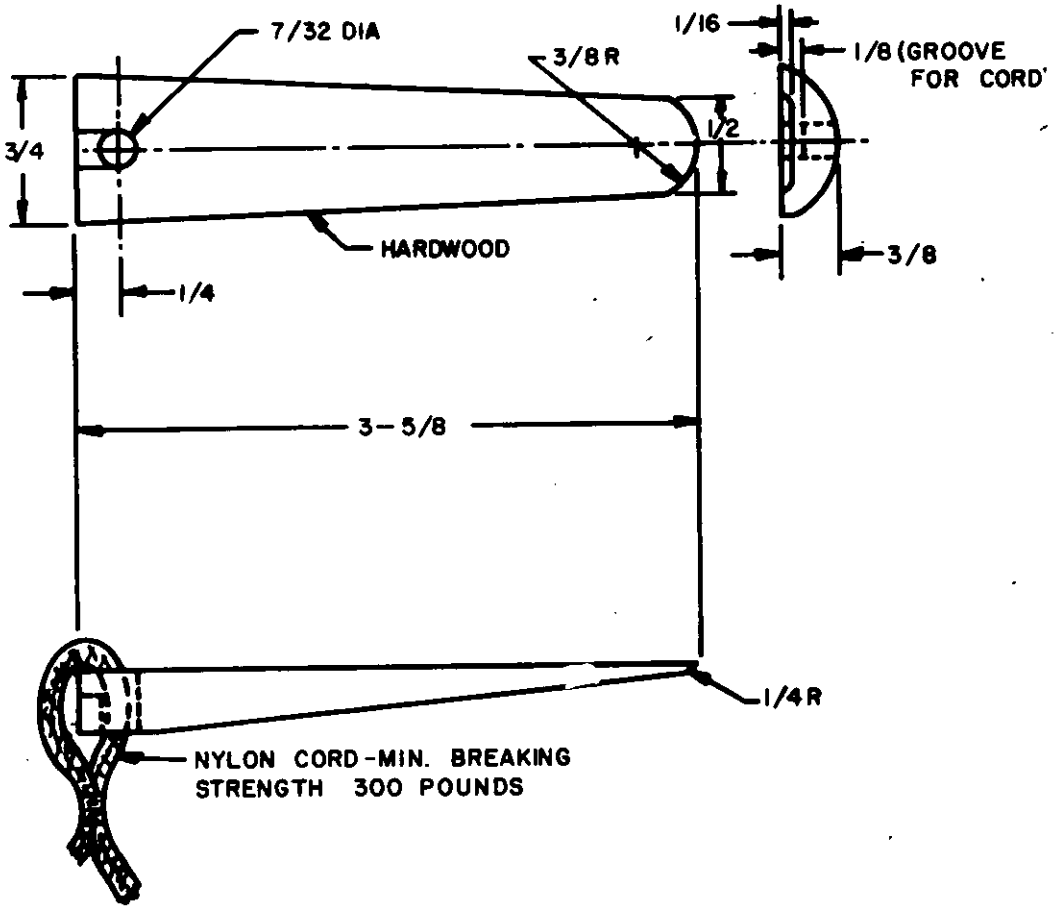


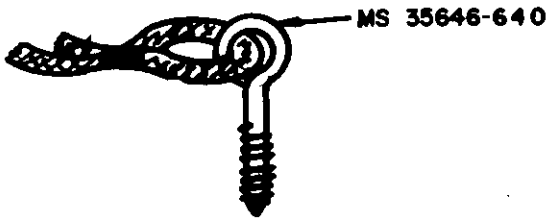
NOTE: ALL DIMENSIONS ARE IN INCHES

FIGURE 5. Circular level vial details.

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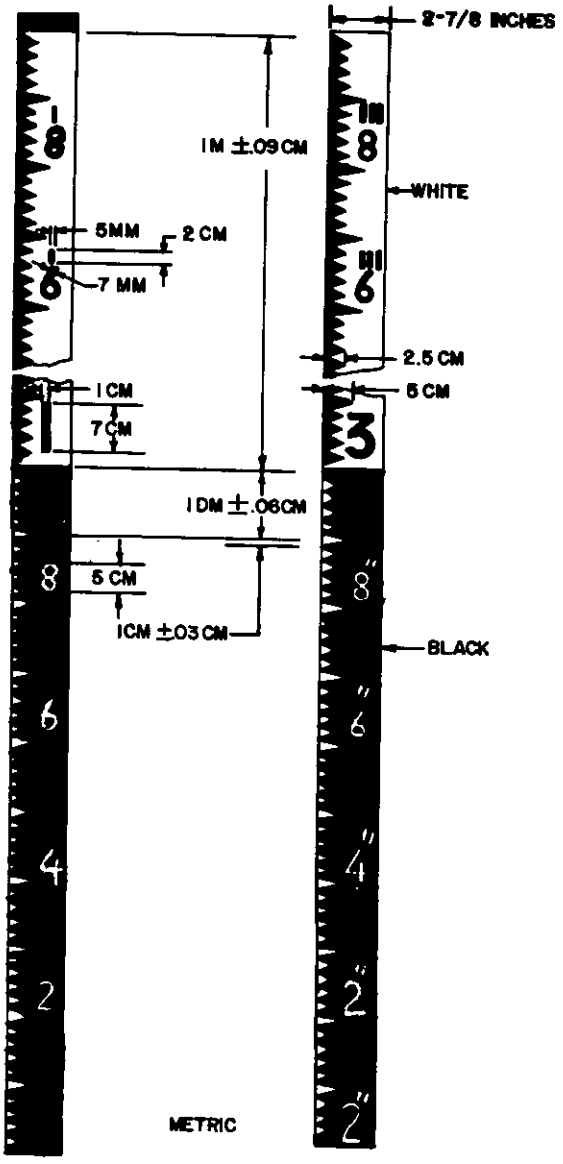


NOTE: ALL DIMENSIONS ARE IN INCHES

FIGURE 6. Locking pin details.

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NOTES:

1. TOLERANCE FOR THE OVERALL LENGTH OF THE GRADUATION AND MARKING SHALL BE .15 CM.
2. ALIGN METER INDICES WITH THE 7 AND 9 CENTIMETER GRADUATION
3. PLACE METERS SO THAT BOTTOM OF NUMBERS ARE ALIGNED WITH THE 2 CENTIMETER AND THE TOP WITH THE 9 CENTIMETER
4. PLACE DECIMETERS SO THAT BOTTOM OF NUMBERS ARE ALIGNED WITH THE 1-CENTIMETER AND THE TOP WITH THE 6 CENTIMETER

FIGURE 7. Graduation and marking design.

X-262[A]

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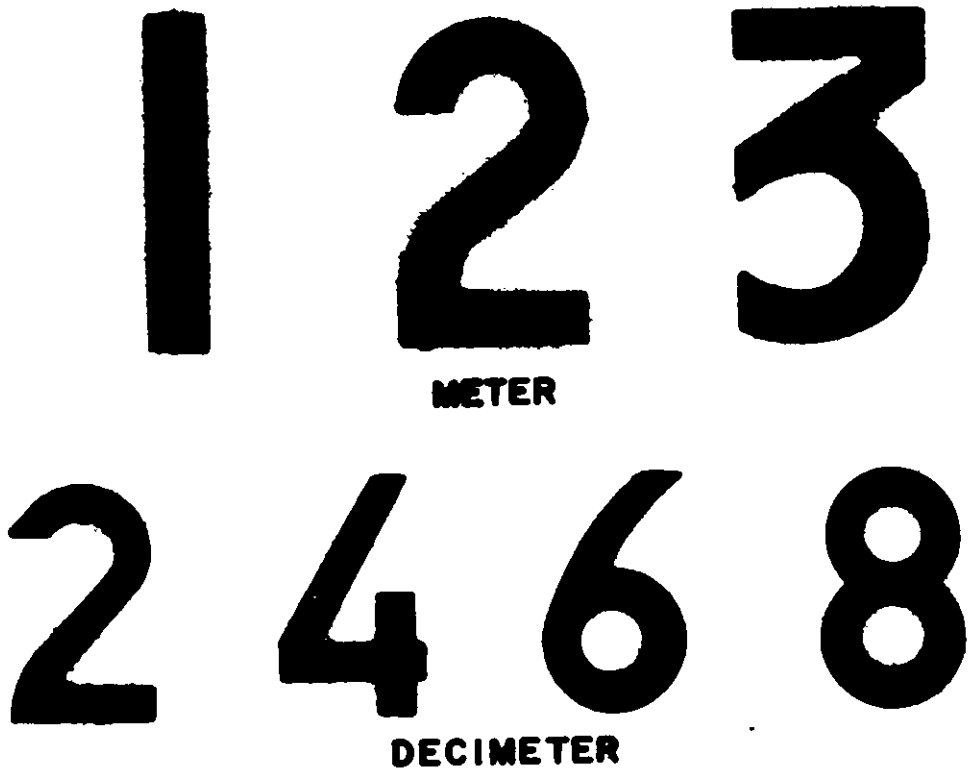


FIGURE 8. Numeral design.

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